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PAPER

Grid Cost and Emissions Externalities of US Data Center Loads with Temoa

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Keywords: energy modeling, grid planning, data centers, emissions,

Abstract

Data centers may add gigawatts of load to the US grid in the coming years. New generation will be necessary to feed their servers and cooling. The type of generation that grid regulators and industry actors decide to build will impact the nation's grid reliability, public electricity costs, and national emissions. We model possible grid futures exploring 'three pillars' regulation that would require data center loads to be met with renewable energy that is (1) additional, (2) hourly-matched, and (3) deliverable. Additional scenarios also explore partial regulation of data centers and uncertainties in future demand, electricity prices, and grid development. These simulations find higher local energy prices and emissions without regulation supporting renewable generation to supply data center loads. If properly regulated, server load growth over future decades can add valuable resiliency to the grid and keep its electricity affordable to all.

1 Introduction

Data center loads made up about 1% of global energy use Cite[masanet,2020] and 2% of US energy use in 2020 Cite [Shehabi, 2024]. By 2024, these servers consumed almost 5% of US electricity generation Cite [Shehabi]. The US Energy Information Agency (EIA) estimates they'll consume between 8% and 10% of US Electricity in 2050 Cite [EIA 2026]. This large electric load requires large infrastructure investments. Industry reports see 40% higher annual utility investments in 2026 compared to 2020 Cite [EEI 2025], with server's exponential growth in a time of grid stress as one reason. In the last PJM capacity auction, new server loads have comprised 40% of the total cost, and the capacity auction fell 6 GW short of their reliability goal Cite [Holand, 2026]. On the ground, data center perception struggles as about $130B across 75 projects have been blocked or delayed in 2026 Cite [data center watch, 2026], and dozens of municipalities across various states have passed bans on new data center construction Cite [Route Fifty, 2026]. AI does has some benefits, from Alpha Fold Cite [Abramson, 2024; Nobel Prize, 2024] to improved grid optimization Cite [Baker, 2025; Owerko, 2020]. Nevertheless, since data centers impact the grid, environment, and the lives around these systems, regulation can prevent server growth from "moving too fast, and breaking things".

One method of regulation may be the three pillars. Deployed in the US, Europe, and partially in Canda, for regulating and alloting tax credits to green hydrogen Cite [Federal Register, 2025; EUR-lex, 2023; Bartlett, 2024]. There's international academic literature already exploring why each of the three pillars is needed to control emissions of electrolytic hydrogen Cite [Ricks, 2023; Zeyen, 2024]. Server grid siting Cite [Vabson, 2026], regulation Cite [Park, 2026; Senga, 2026], emissions Cite [Johnson, 2026] and derived health impacts have also been explored in academic literature Cite [Mueller, 2026]. Moreover, regulators have already mobilized to regulate server loads. E3 notes dozens of large load tariffs in response to server loads weighting down the public grid Cite [Beer, 2026], the Ratepayer Protection Act passed the U.S. House of Representatives this September Cite [Howland, 2026 (2)], and various industry, and regulatory, actors are currently exploring how to manage these new sizeable data center loads Cite [Viswanath, 2026].

Our contribution is evaluating a number of regulatory policies for the regulation of new server loads. To the best of our knowledge, we are the first work to evaluate policy impacts on server load

adoption throughout the continuous US. We do this at an hourly resolution and for 26 regions defined by ISO/RTO boundaries and NERC assessment areas. We produce national and per-region estimates for costs, emissions, capacity expansion and generation profile impacts of explored policies.

2 Results

All results below come from the CONUS 26-zone model at 3,360 hourly time slices per year, solved myopically over 2027–2040. "Coverage" is the share of each zone's hourly server load that must be met under the three pillars; the rest of the server load is served by the ordinary grid. "New-Build 3P" applies additionality as a new-build (vintage $\geq$ 2027) filter only; "Forecast 3P" adds a counterfactual-additionality floor (Methods). Present-value costs are discounted to 2027 at 2% and corrected for the myopic capital-cost truncation described in Methods; no correction is applied to the solver itself. NYCW's server side is excluded throughout because its server load is met by the unserved-load slack.

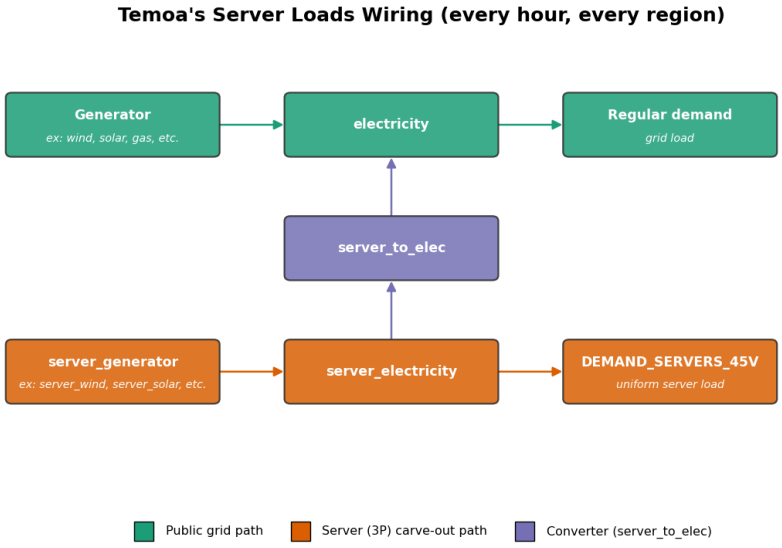


Figure 1. How server load is wired in Temoa, in every hour and every region. Public generators feed the `electricity` commodity, which serves the regular grid load. Eligible clean generators built for servers feed a separate `server_electricity` commodity, which serves the regulated server load (`DEMAND_SERVERS_45V`). A one-way converter (`server_to_elec`) lets surplus server electricity spill into the public grid; nothing flows the other way.

Figure 1 shows the carve-out that every three-pillars result relies on. Because server electricity is its own commodity, Temoa's per-slice balance enforces hourly matching on it, and leaving out a transmission clone keeps it in its zone. The converter is the only link between the two networks: carve-out generation can serve other load when it exceeds the servers' needs, but grid electricity cannot count toward the covered share.

Figure 2 shows the build-out behind each scenario. Adding server load with no regulation (0%) raises 2040 capacity from 1,732 GW (no servers) to 1,893 GW. Raising coverage from 20% to 100% adds a further 139 GW net, from 1,897 to 2,036 GW, and lifts battery capacity from 135 to 212 GW. Most server-dedicated wind and solar (hatched) takes the place of public renewable build rather than adding to it. Forecast 3P builds the most (2,187 GW). Gas alternatives at 100% build less in total (1,871 GW with combustion turbines, 1,848 GW with combined cycles), because firm gas capacity replaces the overbuild of variable renewables and storage that hourly matching otherwise needs. Relaxing hourly matching to an annual balance (p1p3) cuts the total back to 1,904 GW, close to the unregulated case.

Total 2040 generation hardly moves with coverage (6,494 TWh at 0% to 6,527 TWh at 100%), since the load is the same. What changes is the mix (figure 3). Server-dedicated wind and solar grow with coverage and mostly replace public wind and solar. Public gas generation falls by a few hundred TWh between 0% and 100%, and coal barely changes. Spillover to the public grid grows from roughly 400 TWh at 20% coverage to about 850 TWh at 100%. Hourly matching sizes the carve-out for its hardest hours, so in most other hours it produces more than the servers use and the surplus serves other load. Under the gas options, the hatched segment is gas generation instead.

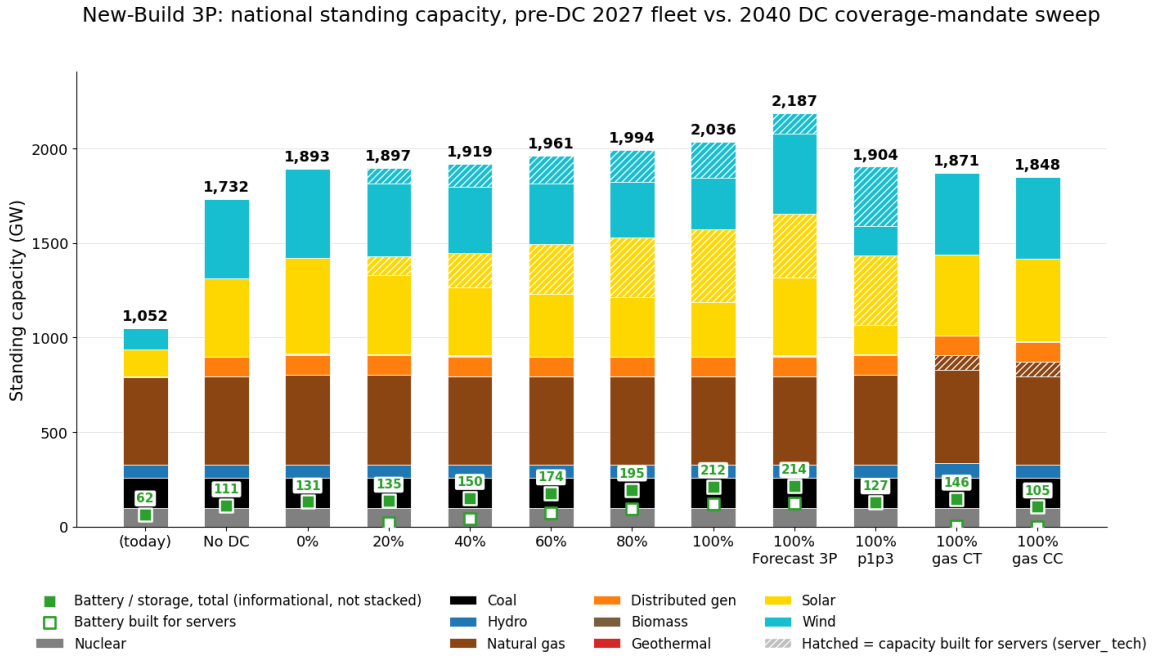


Figure 2. New-Build 3P national standing capacity in 2040 (GW) by technology family, from the pre-2027 fleet through the coverage sweep. Solid bars are public build; hatched bars are capacity built for servers. Green squares mark total battery capacity (not stacked); hollow squares mark the battery capacity built for servers. "p1p3" is 100% coverage with hourly matching relaxed to annual matching.

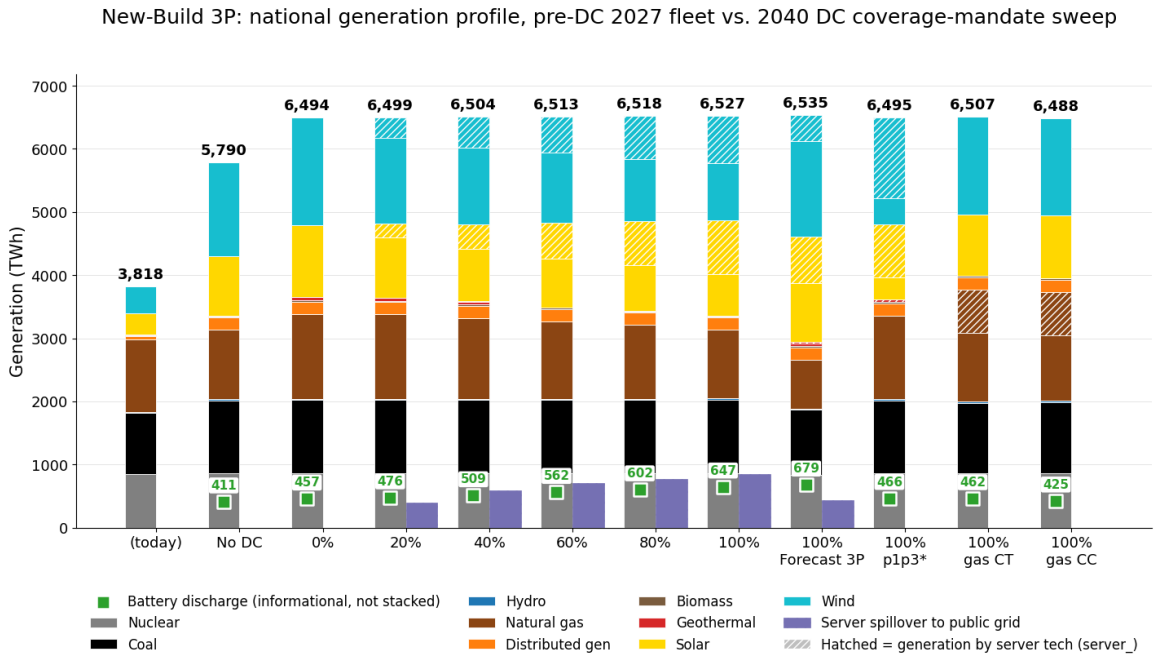


Figure 3. New-Build 3P national generation in 2040 (TWh) by technology family. Hatched segments are generation by server technologies. The purple bar beside each stack is server electricity spilled to the public grid; it is already counted in the stack and is drawn separately so it is not double-counted. Green squares mark battery discharge (not stacked). The p1p3 run has no equivalent spillover mechanism.

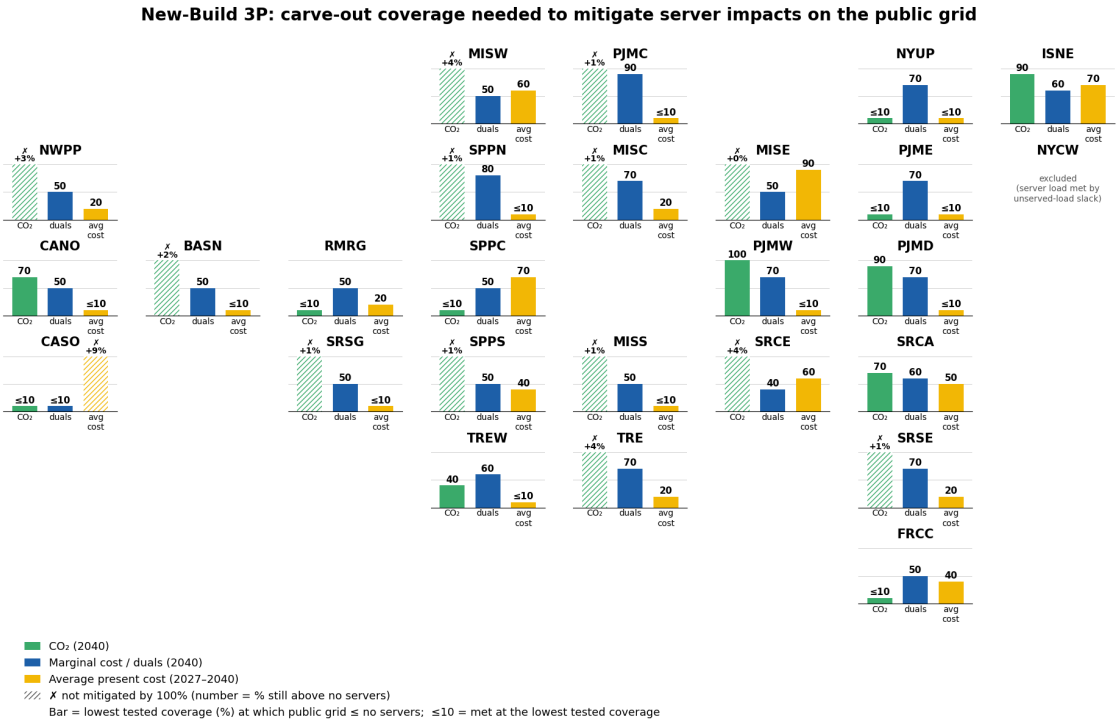


Figure 4. New-Build 3P, by zone: the coverage needed to mitigate server impacts on the public grid, i.e. the lowest tested coverage (10–100%) at which the public grid is at or below the no-servers case for 2040 CO_2 (green), the 2040 demand-weighted marginal cost (dual of the demand balance, divided by the 2040 period weight; blue), and the 2027–2040 average present cost of energy (yellow). A bar labelled ≤ 10 means the threshold was met at the lowest tested coverage. Hatched full-height bars with a cross mark impacts not mitigated by 100% coverage; the number is how far above. The average cost includes the myopic capital-cost truncation correction.

Figure 4 asks, zone by zone, how much of the server load must be covered to mitigate the servers' impact on the public grid, taking the no-servers case as the mark. The three metrics answer differently. Average-cost impacts are mitigated earliest: 11 of 25 zones are back at or below the no-servers case at $\leq 10\%$ coverage, five more by 20%, and only CASO is still above at 100% (+9%). Marginal cost needs much more: 40–90% coverage in every zone except CASO ($\leq 10\%$), mostly 50–70%. Emissions impacts are the hardest to mitigate. Thirteen zones still emit more than in the no-servers case at 100% coverage, by 0–4%. In the other twelve, emissions impacts are mitigated, half of them at $\leq 10\%$.

The gap between the average-cost and marginal-cost thresholds follows from strong duality. At the optimum, total cost equals the sum over constraints of right-hand side times dual, $C^* = \sum_i b_i y_i^*$. Separating the demand-balance term $\Pi = \sum d\,\pi$ (demand priced at the marginal cost π) from the rest and dividing by discounted demand D gives

$$\frac{C^*}{D} - \frac{\Pi}{D} = \frac{1}{D} \sum_{i \notin \text{demand}} b_i y_i^*, \tag{1}$$

so average cost exceeds the average dual exactly when the non-demand constraints contribute positively, and falls below it when they contribute negatively. New-build capacity nets out: its capital cost appears in both C^* and the duals. The contributing constraints split into two types:

- **Push average cost above the duals** ($b_i y_i^* > 0$): binding *floors* ($\geq$ constraints with positive right-hand side and nonnegative dual), such as clean-energy-share requirements and minimum-capacity limits, which force in resources costlier than the marginal one. Existing capacity whose dual is positive does the same: plants that cannot retire and save nothing at the margin, so their fixed O&M is a sunk cost spread over every MWh.
- **Pull average cost below the duals** ($b_i y_i^* < 0$): binding *caps* ($\leq$ constraints with nonpositive dual), such as resource-potential limits, maximum-capacity limits and emission caps. These hold back a resource cheaper than the marginal one and earn it a scarcity rent. Existing capacity whose dual is negative does the same: legacy plants that are cheaper than new marginal build earn an inframarginal rent through avoided capital and fuel costs.

With no existing capacity and no binding limits, average cost and the average dual would coincide. In these runs, covered server load plausibly mitigates the public average-cost impact quickly by spreading existing-fleet fixed costs and the spilled carve-out surplus over more MWh. The 2040 marginal cost, by contrast, stays set by the marginal (mostly gas) unit until enough carve-out supply displaces it in enough hours, which takes far more coverage. The identity explains why the two metrics can diverge, but it is not an exact accounting between the bars here. It holds system-wide and per myopic window for the cost as solved. Our duals are 2040-only, while the average cost spans 2027–2040 and carries the truncation correction. Per zone, interzonal transmission adds congestion rents that the identity assigns only system-wide.

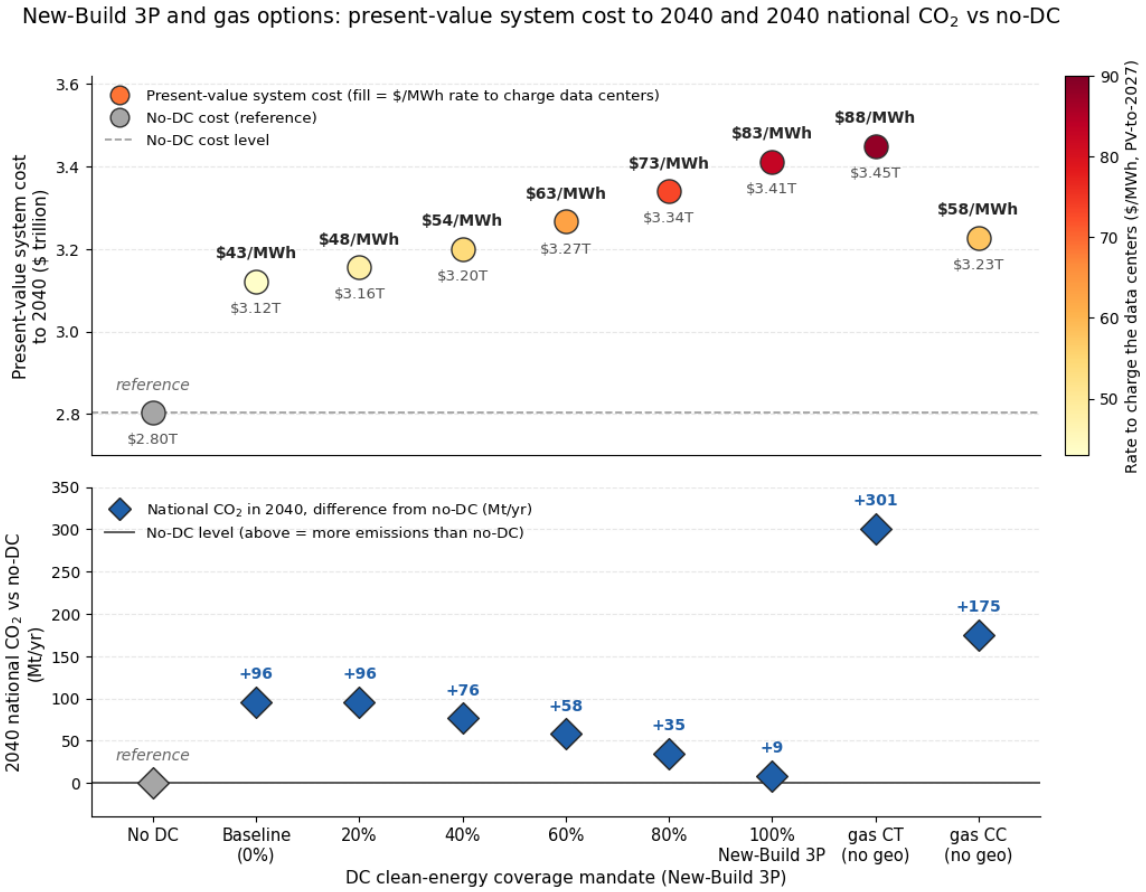


Figure 5. Top: New-Build 3P present-value system cost to 2040 ($ trillion). Each point is labelled with the flat $/MWh rate that, charged to data centers on their whole consumption, would recover all cost above the no-servers case. Bottom: 2040 national CO_2 relative to the no-servers case (Mt/yr). The gas CT and gas CC points meet 100% coverage with new gas plants instead of clean generation (no geothermal).

Serving data centers is expensive in any form: adding server load with no regulation raises present-value system cost from $2.80T to $3.12T, equivalent to $43/MWh of server load (figure 5). Each step of coverage adds cost, reaching $3.41T ($83/MWh) at 100%, and each step cuts emissions. The +96 Mt/yr of 2040 CO_2 added by unregulated servers falls to +76, +58 and +35 Mt/yr at 40, 60 and 80% coverage, and to +9 Mt/yr at 100%. The 20% mandate buys no emissions reduction relative to the unregulated case. The gas options show the trade-off: gas CC is cheaper than 60% coverage ($3.23T, $58/MWh) but adds 175 Mt/yr. Gas CT costs more than 100% clean coverage ($3.45T, $88/MWh) and adds 301 Mt/yr.

At full coverage the public grid pays less per MWh than without servers in 24 of 25 zones (figure 6). The largest savings are in TREW (−10.6), SPPN (−9.5), PJMW (−9.0) and PJMC (−7.8 $/MWh). NYUP (−55.2 $/MWh) is an outlier. CASO is the only zone where the public grid is worse off (+1.9 $/MWh). The server side pays for this: server average cost ranges from $52/MWh (CANO) to $296/MWh (ISNE) and $954/MWh (NYUP).

Figure 7 separates who pays. Unregulated server load raises the public average cost from $38 to $42/MWh. Coverage mitigates this increase: the public average cost is $38/MWh at 20% and falls below the no-servers level beyond that, reaching $33/MWh at 100%. The server side moves the other way, from $72/MWh at 20% coverage to $128/MWh at 100%. The three pillars thus move

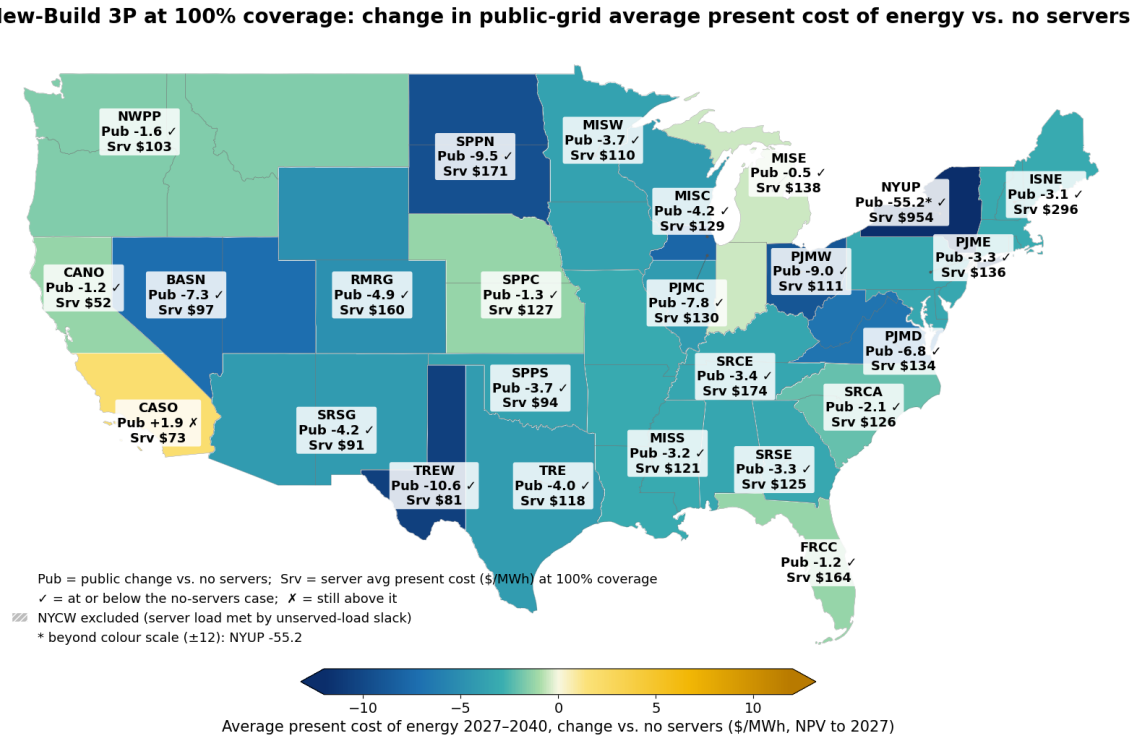


Figure 6. New-Build 3P at 100% coverage: change in the public-grid average present cost of energy (2027–2040, $/MWh, present value to 2027) relative to the no-servers case. “Pub” is that change; “Srv” is the server-side average present cost. A check mark means at or below the no-servers case; a cross means still above. NYUP (−55.2) is beyond the colour scale; NYCW is excluded.

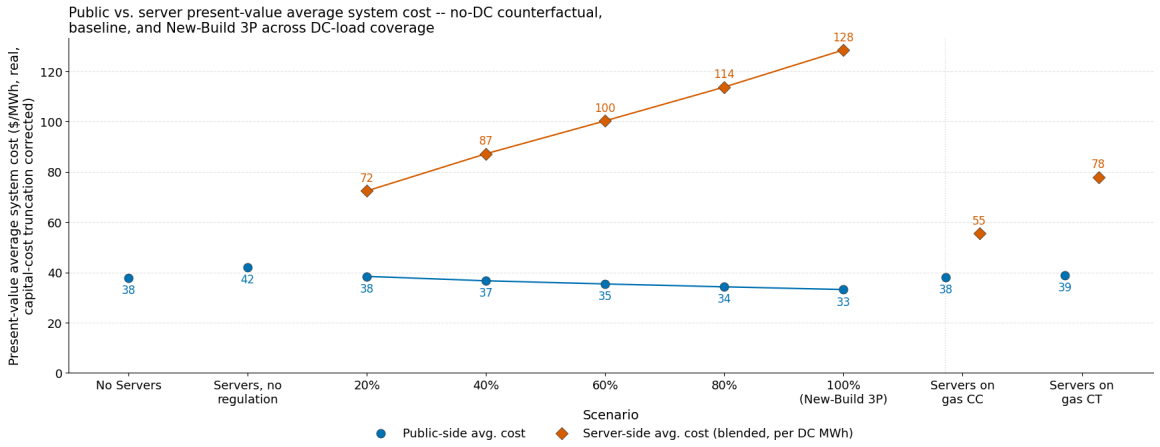


Figure 7. National public-side and server-side present-value average cost ($/MWh) for the no-servers case, servers without regulation, New-Build 3P at 20–100% coverage, and servers served by new gas CC or gas CT. The server-side cost is blended: covered MWh at the carve-out cost plus uncovered MWh at that scenario’s public rate, divided by total server MWh. Server fuel in the gas cases is charged to the server side.

the cost of new server load off the public grid and onto the servers. Gas options keep public costs near the no-servers level ($38–39/MWh) at a lower server cost ($55/MWh CC, $78/MWh CT), but with the emissions shown in figure 5.

Supplementary figures

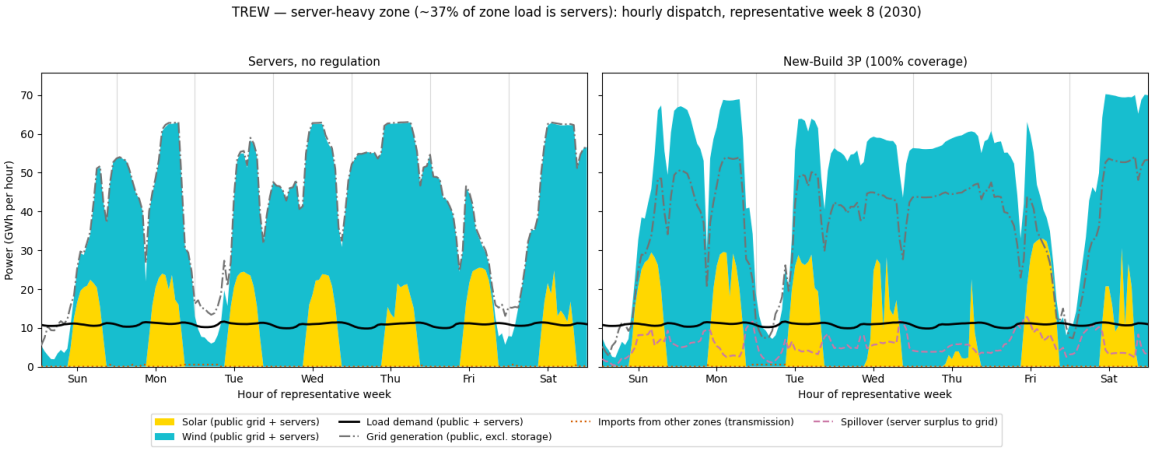


Figure 8. Hourly dispatch in TREW, a server-heavy zone where servers are about 37% of load, for representative week 8 of 2030. Left: servers with no regulation. Right: New-Build 3P at 100% coverage. Stacked areas are solar and wind (public and server combined); lines show load, public grid generation excluding storage, imports, and spillover from the carve-out to the public grid.

Figure 8 shows the spillover mechanism at hourly scale. Under New-Build 3P, TREW's wind and solar output rises well above its unregulated level, and the carve-out spills about 3–12 GW to the public grid in most hours. Public grid generation (excluding spillover) sits lower than in the unregulated case, since the carve-out now supplies part of the zone's output.

Forecast 3P costs more than New-Build 3P at each coverage ($3.25T, $62/MWh at 50% to $3.46T, $89/MWh at 100%) but turns server load into a net emissions reduction (figure 9). Already at 50% coverage, 2040 national CO_2 is 14 Mt/yr below the no-servers case, reaching −260 Mt/yr at 100%. This is consistent with the counterfactual floor's design: it requires carve-out build on top of what the grid would have built anyway, so its surplus cannot simply stand in for public renewables.

3 Discussion

Results vary by region

4 Methods

Temoa is an large scale energy model. Our experiments use the CONUS 26 zone dataset published in may 2025 and available at Cite [OEO, 2026]. We represent the US as 26 zones roughly defined by NERC []. Demand is at an hourly resolution and sourced from [NREL] and projected to 2040 from [IEA AEO]. Generation, also at hourly resolution is regionally based. Each region has a collection of built and potential energy sources based on [FILL]. Transmisison between regions is lossy and based on the overall energy flow between these regions in the existing grid [CHECK]. The linear program behind temoa sees the demand for a representative 3360 hour (20 week) year and chooses which generation capacity to build, retire, and dispatch to fulfill the period in question. We model four periods: 2027-2030, 2030-2035, 2035-2040, and 2040-2045. The objective is minimal present cost.

4.1 Three Pillars in Temoa

We build simulations including every combination of the three pillars.

We model additionality by constraining the electricity that feeds the servers to come from renewable sources built after 2027. We create a special "server electricity" that can only be made by these eligible plants. When relaxing additionality, we do not take a plant's creation year into account.

Hourly-matching is native to temoa; this hourly-resolution energy model enforces that electric demand meets supply at each hour. When we code the server load such that it can only be met by renewables, hourly-matching is automatically enforced. When relaxing hourly-matching, we allow

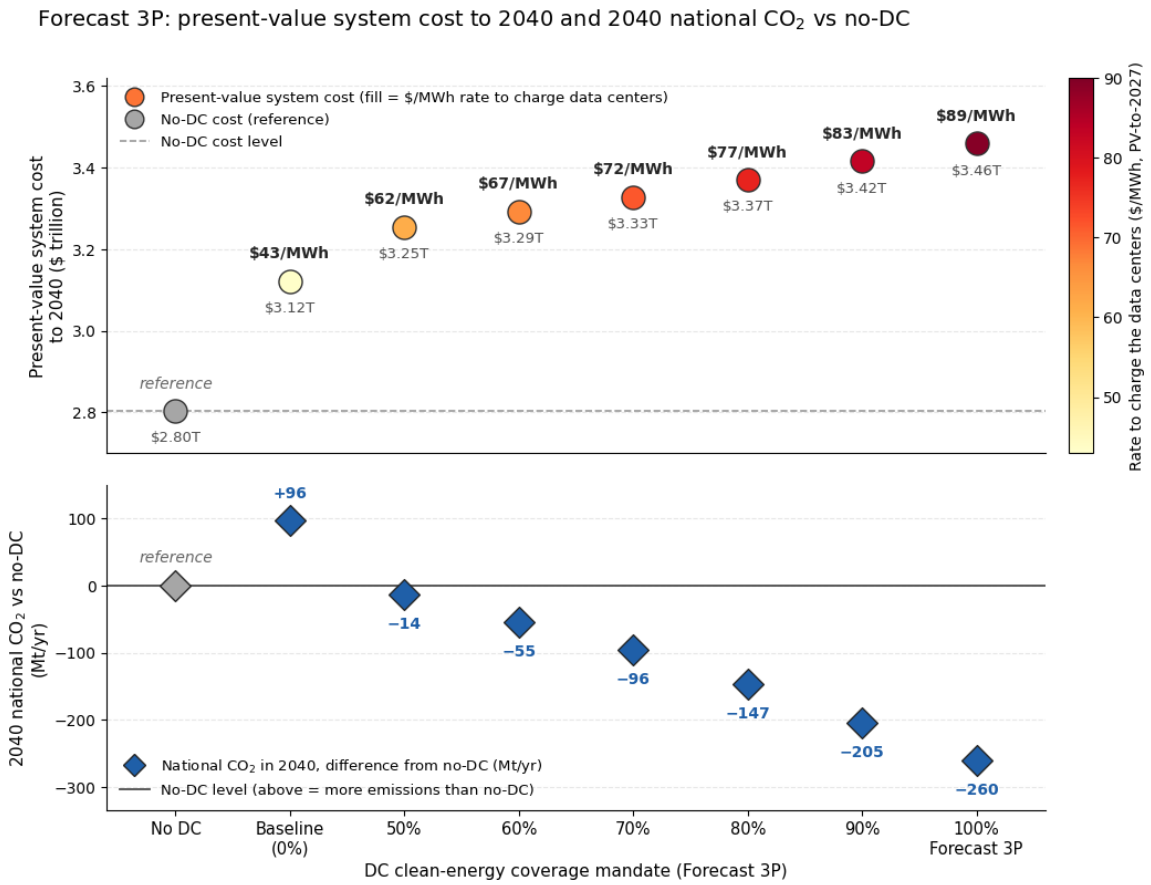


Figure 9. As figure 5, for Forecast 3P (new-build filter plus counterfactual-additionality floor) at 50–100% coverage.

servers to be fed with both server-electricity and regular electricity. If not hourly-matched, we require that the total, annual generated server-electricity from eligible plants equal the total, annual demand of the servers.

We enforce deliverability by forcing the server load to come from generators in its same NERC [CHECK] region. We do not build transmission for server-electricity. When relaxing deliverability, we build server-transmission lines in the model such that eligible server-electricity to feed data centers can move across regions.

We also include a server-electricity to electricity converter. This allow leftover server-electricity to flow into the regular grid and meet the load of other sectors like transportation and industry.

One noteworthy decision here is relaxing hourly matching whenever deliverability is relaxed. This is because without deliverability, energy procurement becomes an accounting exercise rather than a simulation of the grid graph's flows. Hourly matching without deliverability would thus enforce deploying server-electricity across hours with batteries and overbuilt renewables, and doing so up to thousands of miles away from where the load is.

4.1.1 Partial Coverage We model the complete regulation of servers under the three pillars, and we also model regulating various percetages server load at every region and hour. In some cases, it is not necessary to regulate every MWh of server load to keep average costs, marginal costs, and CO2 emissions "unimpacted" from the addition of servers.

5 Conclusions

Hourly matching has the largest impact in the grid's total present cost and emissions and how much generation gets built. Deliverability is also impactful, though the model herein does not capture all transmission congestion issues facing our current and future grid. Additionality, here applied as a rule that energy must come from newly-built generators, does not change total costs, emissions or built generators much. This is because within the eligible generators, there are only a few hundred [CHECK] GWh's available as of the present grid. Whether we enforce new construction of renewable generators or not, they must be built to feed the TWh's of data center

load seen by the grid.

Acknowledgments
Sample text inserted for demonstration.

Funding
Sample text inserted for demonstration.

Author contributions
Sample text inserted for demonstration.

Data availability
Sample text inserted for demonstration.

This energy model of the contiguous United States is available at https://datadocs.openenergyoutlook.org/en/latest/datasets.html.

The code underlying our implementation of three-pillars and its analysis is available at https://github.com/DavidLikesLearning/temoa-dataCenter-3Pillars.

Supplementary data
Sample text inserted for demonstration.

References